

TABLE II
SoCs OF AFFORDABLE SMARTPHONES

System on chip	Manufacturer	Technology	CPU core (no.)	Memory Type	Geekbench 4.0 64 bit single-core score	Geekbench 4.0 64 bit multi-core score
Xiaomi Surge S1	Xiaomi	28nm	4	LPDDR3	678	2416
Samsung Exynos 7870	Samsung	14nm	8	LPDDR3	726	3603
Snapdragon 625	Qualcomm	14nm	8	LPDDR3	864	4165
Hisilicon Kirin 655	Huawei	16nm finFET	8	LPDDR3	824	3401
Mediatek MT 6750	Mediatek	28nm	8	LPDDR3	628	2485
Intel Atom Z2580	Intel	32nm	2	LPDDR2	1046	2515
Spreadtrum SC 9850	spreadtrum	28nm	4	NA	435	1273

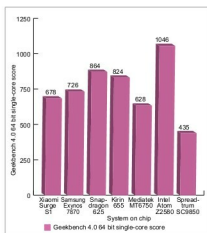


Fig. 3: Single-core CPU processing score for low-end SoCs

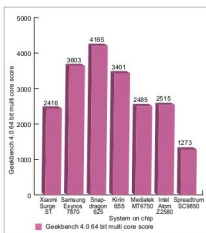


Fig. 4: Multi-core CPU processing score for low-end SoCs

improve thermal management and smartphones' download speed for multimedia applications. Extra features embedded include artificial intelligence/camera connectivity and support for WiFi and Voice-over-LTE (VoLTE) application requirements," says Neeraj Sharma, head of Spreadtrum Communications.

Steve Roddy, senior group director with Cadence Design System, shares that smartphones are going to cover a lot more dimensions than just taking selfies. Now, SoC applications can vary from the Internet of Things (IoT) to drones and security cameras.

Efficient power management.

Smartphone manufacturers target features like high-speed RAM, high-speed processor and multicore processor to offer a customer advan-

tage. As the evolving multimedia services require high-speed data, power optimisation with features such as upgraded RAM, processor speed and multi-core CPUs is important.

Gated bus is a low-power-consumption technology used in SoCs. FinFET technology is being used by

most SoC manufacturers. Qualcomm's Snapdragon 835 uses 10nm technology with FinFET process node. More the fabrication nodes are shrunk, faster the system-on-chip operates and lesser is the power required. The amount of voltage is also lowered due to thinner fabrication node, which has helped in reducing leakage in the chip due to planar processes by up to 50 per cent.

FinFET technology not only cuts down the power consumption but it also enhances the performance. For instance, Qualcomm has been able to reduce the size of Snapdragon 835 by 35 per cent. The chip is also claimed to use 25 per cent less power than previous releases. Thus, the market for low-power-consumption technology is taking a step forward as it impacts the battery life and download speeds too.

Peak download and upload speed. Huawei's Kirin 970 SoC released lately features the maximum download speed (up to 1.2Gbps), while Xiaomi Surge S1 has the minimum speed (150Mbps).

Vulkan API. Vulkan is one of the most important features for high-performance 3D graphics applications—for example, video games and interactive platforms. It is a low-level API that supports parallel tasking, thus reducing the CPU usage. Xiaomi Surge S1, Spreadtrum 9853 and Snapdragon 835 are amongst recently released chipsets that use Vulkan technology. Snapdragon 835 supports Vulkan also to increase multi-core CPU utilisation.

VoLTE technology. VoLTE standard for mobile communication supports

SoCs: Technology and Fabrication

Chips are tested for logical correctness and then sent to the foundry for fabrication. This process is called functional verification. Hardware verification is an important stage as complexity of the design increases with the number of applications. A hardware verification language has been developed in order to test SoCs. So errors, if any, can be identified during the verification process itself.

Simultaneously, hardware elements are collected and passed through the logic synthesis process for performance check related to operating frequency and signal delays. After performance tests, the logical netlist is prepared, following which the initial physical routing is done.

The netlist process helps in analysing the static timing modelling, simulation and other functions which are required to check key parameters such as frequency, power consumption, functional integrity versus electrical integrity, and, above all, power dissipation.

After the verification test of physical layers and performance check, chips are sent to the foundry and later to the wafer fabrication plant to create SoC dies.